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RESEARCH ARTICLE



Quantum neural networks successfully calibrate language models

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Abstract
Recently, quantum neural networks have been applied to different problems, like in computer vision and natural language processing. In this paper, we show an application of quantum models as probability calibrators to language models in Question and Answering (Q&A) tests. Our experiments show comparable results relative to the baseline Q&A models on all metrics. Here, we show another application of this class of models as confidence calibrators, where the parametrized quantum circuit receives a set of features coming from the language model and has to output a corrected confidence measure. The idea is to have a quantum model tell us when the language model is generating good answers or not.

Keywords Quantum machine learning · Natural language processing · Question Answering

1 Introduction and background

Since the great success of deep learning, humanity has definitely entered the age of big data. By providing the right architecture and training data, we can create models that can outperform the best human experts. The possible applications of deep learning techniques are quite broad, reaching several fields of the artificial intelligence. For example, in computer vision (He et al. 2015; Szegedy et al. 2014), natural language processing (Vaswani et al. 2017; Devlin et al. 2018), deep reinforcement learning (Silver et al. 2018; Open et al. 2019) and also automatic speech recognition (Hinton et al. 2012). Despite this huge amount of applications, it is important to emphasize that the high performance of deep neural networks comes at the cost of having poorly calibrated confidence scores or probabilistic overfitting (Guo et al. 2017). The confidence score indicates when an answer is likely to be incorrect. This is a crucial task for many applications, especially high-risk ones as, for example, self-driving cars. In the context of Question and Answering problems, a wrong answer misleads the user. To overcome these barriers, recent manuscripts have explored the calibration of these models to correct their output probabilities. In general, another classifier is used in a post-processing step, which takes as input some tailored features coming from the deep neural models (Jia and Xie 2020; Chakravarti and Sil 2021). The number of tailored features is usually few, in the order of a few dozen, which opens the possibility of using Noisy Intermediate Quantum Scale (NISQ) devices in a post-processing step. This is exactly the strategy that we will follow in this manuscript.

In parallel with the deep learning revolution, a few decades ago, a new field of quantum physics emerged, the quantum computation and the quantum information theory. The Information and computation theory expanded its possibilities, including new ways of transmitting, storing, and processing information. In spite of the considerable growth of the field, processing large amounts of quantum information is still a difficult task. Indeed, the environment interacts with the data, and sustainably blocking undesirable interactions for thousands of qubits is still a pursuit of the scientific community. In contrast to early quantum algorithms, when dealing with NISQ devices, only hundreds of noisy qubits are required. Although universal quantum computing for NISQ devices does not yet exist, specific computing tasks can be operated

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on such platforms. An important example is the method we use in this manuscript, the variational quantum algorithms (VQA) that can deal with practical problems such as combinatorial optimization and machine learning. VQA is built through a properly parameterized operation whose parameterization is obtained by means of a classical computer.

Recently, quantum machine learning has shown its potential advantages by recent works, especially in terms of expressivity (Abbas et al. 2021) and generalization (Huang et al. 2021), which are measures to determine the efficiency of a model. Expressivity, according to Abbas et al. (2021), quantifies how much a variational circuit can access the Hilbert Space. On the other hand, generalization is the capacity of the model to adapt to unseen data.

In this paper, we explore how Parametrized Variational Circuits can be used to distinguish patterns of questions coming from different domains to produce a calibrated score. Note that VQA does not generate the answers, as this is done by the language model, which is classic. Instead, it only calibrates the probabilities that come from the language model itself. After calibrating the scores with the quantum model, incorrect answers with high confidence have a low probability, and thus we can abstain from showing an incorrect answer. Nevertheless, it is important to emphasize that, despite the enhancement present in the quantum model, some classical algorithms still give better results for some metrics, but for future studies, tailored features can be used for an easier data topology for the quantum models, allowing better scores.

This manuscript is organized as follows. In Section 2, we introduce the classical algorithms used to train the Q&A model. Such approaches give the answers that generate features for the confidence calibration. In. Section 3, we present the confidence calibration, a method that returns a confidence score for a given answer. For this purpose, we use three distinct measures for calibration error: the Average Calibration Error, the Maximum Calibration Error, and the Brier score. In addition, we introduce the idea of a reliability plot, which is a simple way to represent the performance of a probabilistic prediction for a binary event. In. Section 4, we present the variational circuit used to develop the confidence calibration using quantum states of quantum computers, and in Section 6, we present out experiments and results. We conclude our manuscript in Section 7

2 Q&A models

Question Answering, also known as Machine Reading Comprehension, is a traditional task where machine learning models learn to find the best answer inside some context (paragraph), given some question. Each answer predicted by the model is a sentence chosen in the paragraph and is commonly referred to as an answer span. In recent years, Deep Learning state-of-the-art in natural language processing had a breakthrough since the introduction of the Bidirectional Encoder Representations from Transformers (BERT) architecture (Devlin et al. 2018). In a nutshell, BERT consists of words and positional embeddings, a series of encoder-decoder tailored layers, and attention heads (Bahdanau et al. 2014; Vaswani et al. 2017), where the latter not only allows the model to capture the semantics of how tokens relate to each other in the given text but have also been used extensively in the interpretability of these neural architectures (Guidotti et al. 2018; Letarte et al. 2018; Vashishth et al. 2019; Clark et al. 2019). It is important to emphasize that this is an often difficult task to perform in deep learning, and it is an active area of research. From an unsupervised training step, it learns to efficiently model the language used for training, which is usually performed from pre-training on a large corpus of publicly available texts such as Wikipedia. This allows the model to learn meaningful latent representations of the textual semantics that can be used later for multiple tasks — like Machine Translation, Text Classification, Information Retrieval, among others — after a fine-tuning step.

The models considered in this paper were fine-tuned on the Stanford Question Answering Dataset (SQuAD2.0) (Rajpurkar et al. 2018), which consists of many paragraphs followed by a list of many questions about Wikipedia articles that were formulated by hired crowdworkers. The answers to these questions are passages contained in paragraphs and some of them are unanswerable and, therefore, do not contain any passages as an answer. Indeed, SQuAD2.0 contains much more unanswerable questions written by the crowdworkers, in an adversarial way, i.e., they are very similar to the answerable ones, but actually contain no correct span inside the paragraphs. In this sense, for a Q&A model to perform well on SQuAD2.0, it needs to know when to refrain from answering.

To study and train a machine learning algorithm based on the dataset above, we use two distinct pre-trained models based on the Bidirectional Encoder Representations from Transformers (BERT) architecture, the *Distilbert-base-uncased-squad2*, and the *RoBERTA-large-squad2* (Liu et al. 2019). This pre-training stage is necessary for the model to learn about the language in a specific task, in this case Q&A. The DistilBERT is a smaller, faster, cheaper, and lighter version of BERT that is obtained by a specific type of training known in the machine learning community as Knowledge Distillation (Gou et al. 2021). In the Knowledge Distillation a *student model* (usually a smaller, lighter, and faster model) distills (learns) the knowledge of a *teacher model* (in this case, the original BERT model). On the other hand, RoBERTA is a BERT model with optimized training parameters. For both, we used the pre-trained version on the SQuad2.0 dataset, but while for DistilBERT we used SQuad2.0 itself for testing, for RoBERTA we used a different dataset, an internal dataset

from NTT Data. Again, we re-emphasize that, in this paper, the language models are not the confidence calibrators. The latter is also a machine learning model, and its objective is to correct the confidence of the answers that are output from the language model. That is the final objective of our work: to show that parametrized circuits can be used in this calibration step.

3 Confidence calibration

Before diving into their mathematical definition, one distinction has to be made: the difference between the model's internal probability score and the observed, or empirical, accuracy. Given a set of n predictions with a score of 90%, for example, in a perfectly calibrated model, one should expect to get about 90% of those n predictions right. However, there are cases where a model is over-confident in the answers, suggesting a score of 90%, while in practice it gets, for example, 70% of correct answers.

For classical computers, in the context of Question Answering, several previous studies have shown detailed methods for correctly calibrating Q&A model scores. For example, in Jia and Xie (2020); Chakravarti and Sil (2021), an XGBoost (Chen and Guestrin 2016) is proposed as a post-processing step to output the corrected probabilities, considering specific and tailored features depending on the neural models. Here, we use as input to our calibrator, a set of features obtained from the neural Q&A model.

In the context of Question Answering, specifically, BERT-based architectures receive a sequence of questions, context (paragraph), and special tokens (N) as inputs. As outputs, we obtain the probability of beginning (α_b) and end (α_e) of the answer spans that are calculated internally. Also, a probability of non-answerability is output as α_{CLS}. Therefore, we use a set of five features: (i) probability of beginning of answer span (α_b), (ii) probability of end of answer span (α_e), (iii) number of words in the question (length of question tokens), (iv) number of words in the answer (length of answer tokens) and (v) no-answer likelihood α_{CLS}. The next diagram illustrates the whole process:

Input Text: "In what year was the Declaration of Independence signed?"
Context: "The United States Declaration of Independence was signed in the year 1776."
BERT Answer: "1776"
(i) $\alpha_b = 1.0$
(ii) $\alpha_e = 1.0$
(iii) number of words in question $= 9$.
(iv) number of words in answer $= 1$.
(v) $\alpha_{CLS} = 0.0$

After the post-processing step, whether by classical or quantum methods, we obtain a bidimensional vector $\mathbf{z}_m athbfi$ as answer: the so-called logits. In the quantum case $\sum_k z_k = 1$ which is not generally true when we use classical algorithms. Thus, as standard procedure, the final probability is obtained by the well-known softmax activation function:

$$\hat{p}_i = \max_i \left(\frac{e^{\mathbf{z}_i}}{\sum_{j=1}^{k} e^{\mathbf{z}_j}} \right) = \max_k \sigma_{\text{SM}} \left(\mathbf{z}_i \right)^{(k)}, \tag{1}$$

where $\hat{p}_i$ is the calibrated confidence for the prediction, i.e, the probability of being a correct or incorrect response, z_i is the output of the machine learning algorithm, be it classical or quantum, k is the number of classes (in our case two, correct or incorrect answer), and σ_{SM} is the softmax activation function.

In order to understand how poorly calibrated a quantum model can be, some measures of calibration error must be considered. To capture this notion of miscalibration, we use in this study three measures (Naeini et al. 2015): the average calibration error (ACE), the maximum calibration error (MCE), and the Brier score. Below we briefly explain the definition of each measure and how we analyze our results.

3.1 Measures of calibration error

To better visualize and understand how these measures of calibration vary, given the value of $\hat{p}_i$, we divide the confidence interval into M equally-spaced bins. Let B_m represent the set of samples for which the predictions fall within the confidence interval $(\frac{m-1}{M}, \frac{m}{M}]$, with m given by an integer between 1 and M. Now, we can calculate the accuracy of each bin B_m as the number of samples with correct prediction within the bin divided by the number of total samples in the bin:

$$acc\left(B_m\right) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbf{1}\left(\hat{y}_i = y_i\right). \tag{2}$$

Similarly, the average confidence of bin B_m is given as follows:

$$conf\,(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}_i, \tag{3}$$

which, as the name suggests, is the average confidence of each interval. It is important to note that, from these definitions, for a perfectly calibrated model, its corresponding diagram should be the identity function, as the average accuracy equals the average confidence, for each bin.

Given the definitions above, the Average Calibration Error (ACE) is formally defined as follows:

$$ACE = \sum_{m=1}^{M} \frac{|B_m|}{n} |acc(B_m) - conf(B_m)|, \tag{4}$$

where n is the total number of samples considered. Intuitively, the ACE measures the average gap of miscalibration of the model. The Maximum Calibration Error, in turn, measures the bin that contains the highest divergence between the empirical accuracy and the average confidence. Indeed, in a handful of applications, one might be interested in minimizing the risk for the worst-case scenario. For this, one should consider the MCE, given by:

$$MCE = \max_{m \epsilon \{1,2,\ldots,M\}} |acc(B_m) - conf(B_m)|. \tag{5}$$

Finally, the Brier Score (BS) measures the accuracy of probabilistic predictions by considering the mean squared error of the outputted probabilities. A score of 0 indicates perfect probabilistic accuracy, while 1 represents the worst scenario. Formally:

$$BS = \frac{1}{n} \sum_{i=1}^{n} (p_i - o_i)^2, \tag{6}$$

where n is the number of observations, p_i is the confidence of the prediction i, and o_i is the actual outcome of i.

4 Quantum variational circuits

Variational circuits, also known as "parameterized quantum circuits," are quantum algorithms that depend on free parameters. Most of the hybrid quantum-classical algorithms rely on these circuits. The key idea of the variational circuit is to optimize those parameters according to a cost function (McClean et al. 2016). Like standard quantum circuits, they consist of (a) the preparation of a fixed initial state, (b) a quantum circuit parametrized by a set of free parameters θ, and (c) the measurement of an observable at the output, that may be made up from local observables for each or a subset of wires in the circuit.

We describe the general structure of the variational quantum circuit used in this manuscript in Fig. 1. The $U(x)$ is the data encoding block which is predefined and static. There are three main ways (Schuld and Petruccione 2018) to encode classical data in a quantum state: directly into the elements of the basis states, in the amplitude of the quantum state, or in the angle of the quantum state.

Here we use the latter and more details about this encoding can be found in Schuld and Petruccione (2018). In our scheme, we use 5 qubits, one encoding each feature. The circuit delimited with the dashed rectangle represents the tunable part, which can either be optimized by a gradient-based (Watabe et al. 2019; Stokes et al. 2020; Sweke et al. 2020; Wiersema and Killoran 2020; Schuld et al. 2019) or a gradient-free method (Franken et al. 2020). These parameters can be seen as the weights in the classical neural networks and in this work we use gradient descent to optimize θ_k with $k = \{1, 5\}$. Finally, the algorithm finishes with a measurement on qubit 1 which returns a binary vector z_i.

In our experiments, we tested two variational circuits with the exact same architecture, except for the measurement stage. In the first model, which we call Raw Quantum Neural Network, only the first qubit is measured as the calibrated probability, resulting in an input to the sigmoid activation function. In the second architecture, we measure all qubits to obtain five probabilities. These results are then

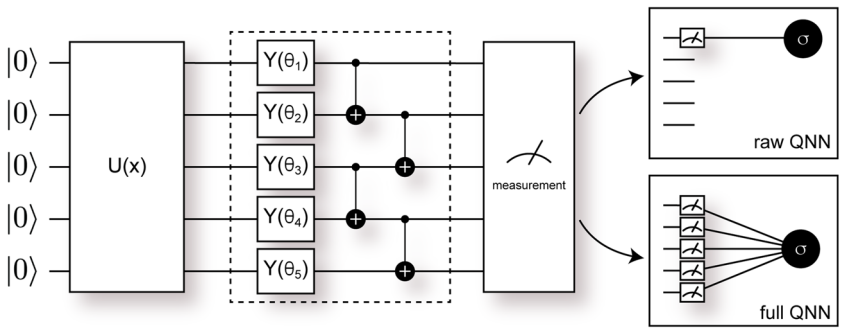


Fig. 1 Quantum variational circuit architecture Chen et al. (2020). $U(x)$ is the quantum routine for encoding classical data, and the circuit delimited with the dashed rectangle represents the variational circuit block with tunable parameters θ_k. At the end, a measurement is made depending on the strategy, raw QNN or full QNN

subjected to a classical neuron with a sigmoid activation function, where new parameters are adjusted. This also outputs the calibrated probability. It is important to emphasize that successful experiments have shown that these circuits have multiple applications, such as function approximation (Mitarai et al. 2018), classification (Havlíček et al. 2019; Farhi and Neven 2018; Benedetti et al. 2019), generative modeling (Lloyd and Weedbrook 2018; Chakrabarti et al. 2019), deep reinforcement learning (Chen et al. 2020) and transfer learning (Mari et al. 2020). Here, we show how it can be used as a calibrator for classical models. Based on previous works of confidence calibration for Machine Reading Comprehension using classical models (Jia and Xie 2020; Chakravarti and Sil 2021), we train our quantum circuit to produce a calibrated score.

5 Implementation

The source code was implemented in Python 3.8, with Pennylane. In our experiments, we trained the variational classifiers for 10 epochs, i.e., the number of rounds used for training; the learning rate ϵ (the step at which the optimization surface is navigated) was set to 10^{-2} and, after 5 epochs, it is set to 10^{-3}. On the SQuAD2.0 dataset, we took a sample of 1000 random questions, where a split of 50%/50% was used for training and testing. On the NTT dataset, of 1159 questions, we used approximately 23% for testing (270 random questions) and the remaining for training. For training, we use a batch size of 64 samples, i.e., the number of questions processed by the model at a time and, in the end, after processing the dataset, an average loss is calculated. The loss function was the Binary Cross Entropy (Painsky and Wornell 2018), and we used the Adam optimizer (Kingma et al. 2020) to determine the minimum of the loss function. These parameters are summarized in Table 1.

6 Experiments and results

To illustrate our results, we use the well-known Receiver Operating Characteristic (ROC) curves, a graph that illustrates the diagnostic capability of a binary classifier system.

Table 1 Hyperparameters used in the calibration experiments

Parameter	Value
Learning rate (ϵ)	10^{-2} and 10^{-3}
Epochs	10
Batch size	64

Table 2 Results for the SQuAD2.0 dataset (best scores are highlighted)

Model	Brier score (↓)	ECE (↓)	MCE (↓)	AUC (↑)
distilbert-base-uncased-squad2	0.4053	0.6385	0.4589	0.52
Raw quantum	0.1993	0.1719	0.1763	0.84
Full quantum	0.1467	**0.1474**	**0.1007**	0.87
XGBoost	**0.1400**	0.1542	0.1055	**0.88**

In a binary classification, there are four possible occurrences in a prediction: the true positive and the true negative, when the predictions are correct, and the false positive and the false negative, which are considered errors. The ROC curve is a graphical plot of the true positive rate against the false positive rate. It gives the ratio between true alarms versus false alarms. Moreover, we use also reliability diagrams (DeGroot and Fienberg 1983; Niculescu-Mizil and Caruana 2005) which allow for a visual inspection of uncalibrated predictions. The reliability plots are computed by plotting the empirical accuracy as a function of confidence.

As aforementioned, we use the average calibration error, the maximum calibration error, and the Brier score as measures of calibration error and, additionally, we also calculate the area under the ROC curve (AUC). The latter indicates how the model is better than random guessing since, the higher this measure, the better.

We can see in Tables 2 and 3 the comparison between the baseline models, *distilbert-base-uncased-squad2* and *xlm-roberta-large-squad2*, and the respective calibrators, for each dataset. For SQuAD2.0, the hybrid variational circuit achieves the best ECE and MCE, while the XGBoost has the best Brier score and AUC. As we can note, all quantum models achieve significant improvements relative to the raw model which has no confidence calibration.

For the internal dataset, the hybrid model achieves the lowest Brier score, while the raw quantum has the lowest MCE. The XGBoost presents the lowest ECE and AUC. Again, all models significantly improve all metrics when compared with the uncalibrated *xlm-roberta-large-squad2 model*.

Table 3 Results for NTT dataset (best scores are highlighted)

Model	Brier score (↓)	ECE (↓)	MCE (↓)	AUC (↑)
xlm-roberta-large-squad2	0.4611	0.5316	0.7691	0.51
Raw quantum	0.1833	0.0497	**0.0916**	0.69
Full quantum	**0.1571**	0.0548	0.1271	0.73
XGBoost	0.1587	**0.0455**	0.0979	**0.75**

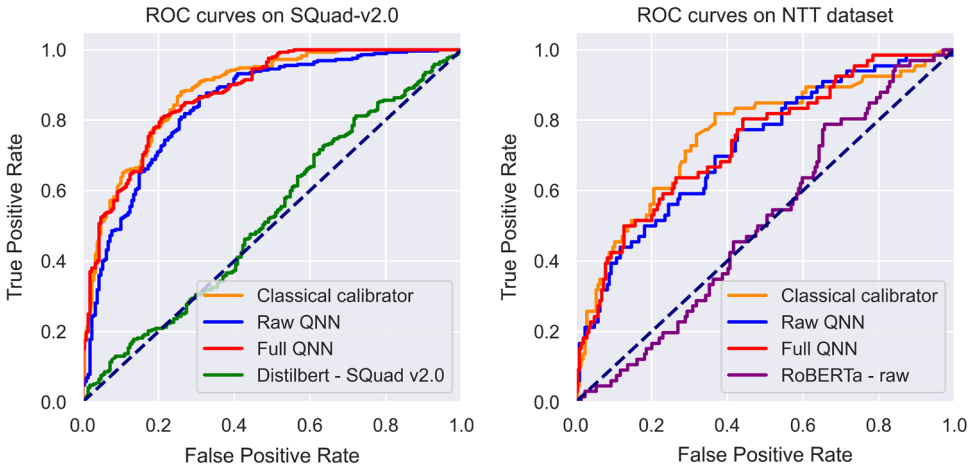


Fig. 2 Squadv2 and NTT dataset ROC curves for baseline models (distilbert-base-uncased-squad2 and xlm-roberta-large-squad2, respectively) and calibrators (XGBoost for the classical calibrator, and raw/full Quantum Neural architectures, as explained previously)

Regarding the AUC scores, we also show the ROC curves comparison in Fig. 2. From the figure, we can see that using the pure probability coming from the Q&A models is not a good confidence measure. As we can see, the uncalibrated model is close to the identity line, which means that a prediction is as good as a coin toss. Also, we note that this is mitigated by using the calibration strategies proposed.

Finally, we present the reliability diagrams for the two datasets (Figs. 3 and 4). The red bars represent the gap between the measured empirical accuracy and the confidence score in each bin, which gives a sense of how uncalibrated the predictions are at each interval of confidence. Blue bars below the identity line indicate over-confident predictions of the Q&A model, where it would have been better to abstain from answering. Above the identity line indicate situations of under-confident answers. This is also undesirable, as we want correct predictions to have a higher probability. The uncalibrated models on SQuAD2.0 dataset and NTT dataset are shown in Fig. 3a and b, respectively, while the calibrators are illustrated in Fig. 4. In both datasets, we can see in Fig. 3 that the gaps are higher when compared to the calibrated version in Fig. 4. Indeed, both the quantum and the classical models have much better scores relative to the respective baselines. It is important to note that bins missing below 50% threshold do not represent uncalibrated predictions, since after the calibration step, there is no distribution of predictions in those bins at all.

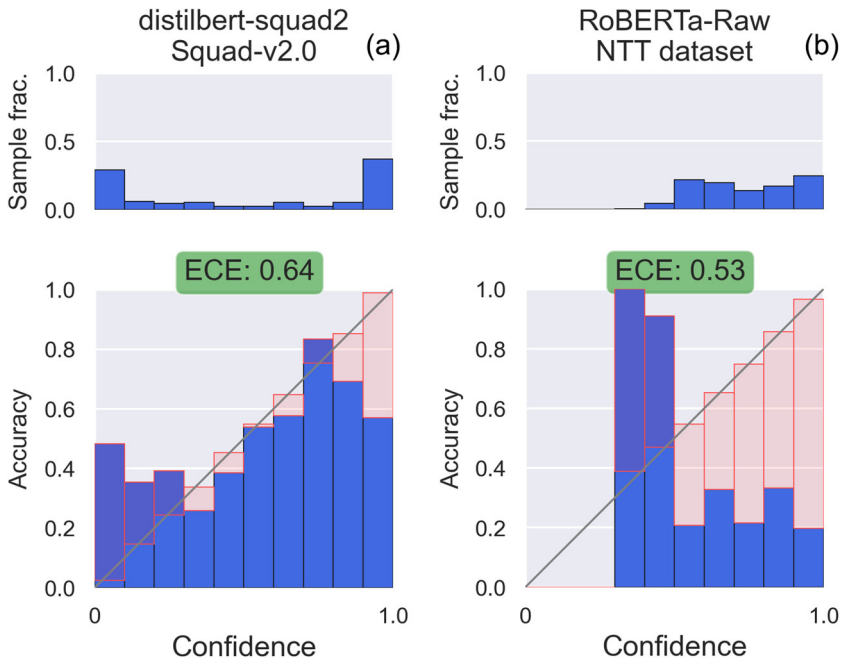


Fig. 3 Reliability plots for both models without calibration. In **a**, we present the results for the SQuAd2.0 dataset, and in **b**, for the NTT dataset

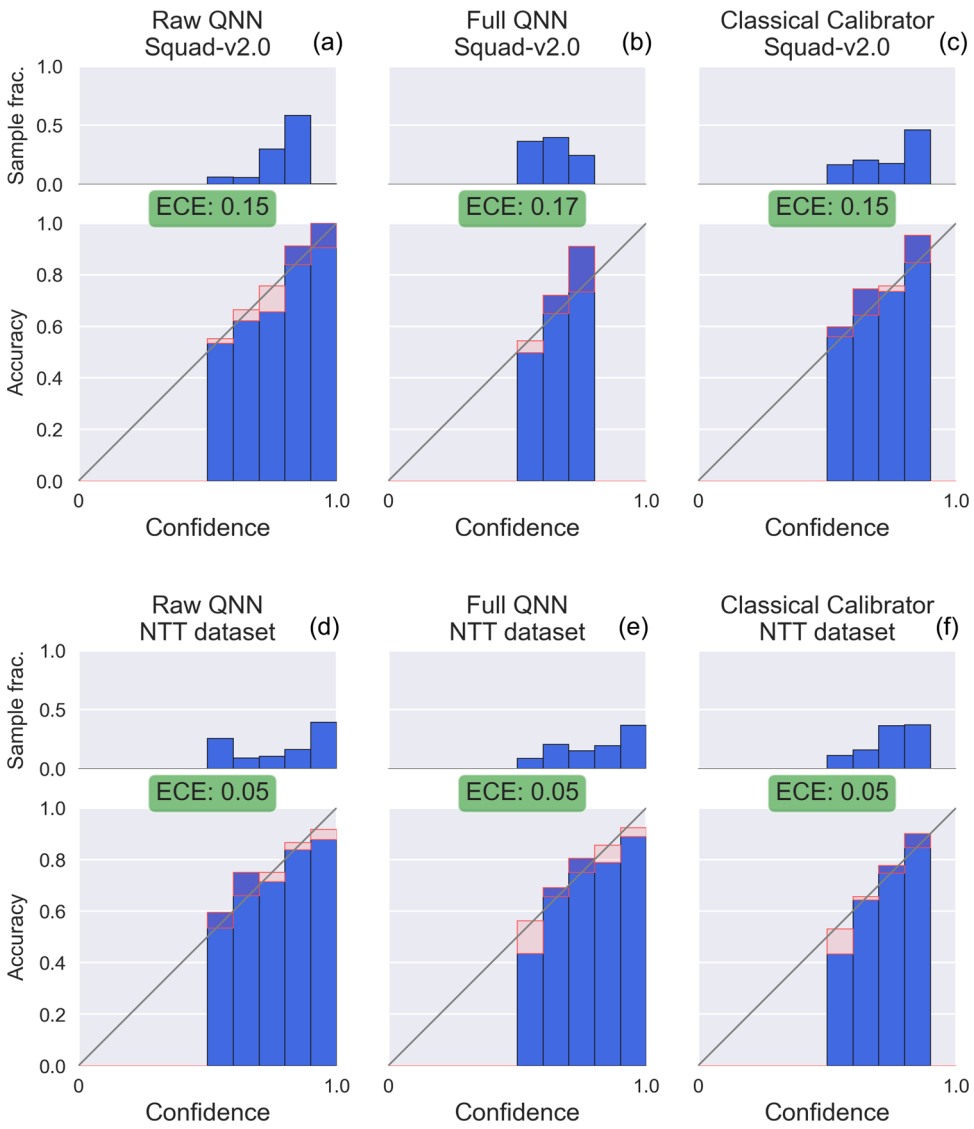


Fig. 4 Reliability plots for both models with calibration. In **a**, **b**, and **c**, we present the results for the SQuAd2.0 dataset using Raw QNN, Full QNN, and the classical calibrator, respectively. In **d**, **e**, **f** is the analogous for the NTT dataset, respectively

7 Conclusion

In this paper, we explored a new application of quantum neural networks, showing that parametrized variational circuits can successfully calibrate Deep Learning Q&A models. The experiments were performed in SQuAD2.0, and an internal dataset from NTT. The results showed significant improvements in all metrics compared to the baseline. We re-emphasize that the improvement is not over the classical calibrator model, but rather it is over the (uncalibrated) probabilities of the language model. We saw how, ultimately, the calibrators (both quantum/classical) are distinguishing patterns of questions coming from different distributions to produce a corrected score. Since we now know that some dataset geometric distributions yield better generalization in physical quantum models, it can be the case that, in the future, we have tailored features that better separate the question domains, allowing for even better-calibrated scores.

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Author contribution JSN conducted the experiments with the classical and quantum machine learning models, generated the results, figures and tables. FFF, JPP, and RCG developed the theoretical ideas, guided the creation process, and coordinated the project. LQ elaborated on the mathematical part of the calibration measures. FFF, JSN, and FA wrote the section about the variational quantum circuits. TNN contributed by explaining the Q&A models. All authors reviewed the manuscript.

Availability of data and materials The SQuAD 2.0 dataset can be found at https://rajpurkar.github.io/SQuAD-explorer/.

Declarations

Conflict of interest The authors declare no competing interests.

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